

MAGRILAN 20MG CAPSULES

(Fluoxetine Hydrochloride)

PHARMACEUTICAL FORM

Yellow/violet cap/body hard gelatin capsule of size '4' containing white free flowing powder
Each "MAGRILAN" 20mg capsule, contains Fluoxetine Hydrochloride equivalent to 20mg of Fluoxetine base
For excipients see section Pharmaceutical Particulars
Capsules for oral administration

THERAPEUTIC INDICATIONS

Depression: MAGRILAN is indicated for the treatment of the symptoms of depressive illness, with or without associated anxiety symptoms.

Obsessive-compulsive disorder

Pre-menstrual Dysphoric Disorder (PMDD): MAGRILAN is indicated for the treatment of pre-menstrual dysphoric disorder. Diagnosis of PMDD: The essential diagnostic features of PMDD are clear and established cyclicality (occurring during the last week of the luteal phase in most menstrual cycles) of symptoms such as depressed mood, anxiety, affective lability, accompanied by impairment in social and/or occupational function and physical symptoms (such as breast tenderness or swelling, headaches, joint or muscle pain, a sensation of bloating, weight gain) – all of which must be severe. This syndrome should be distinguished from the commoner "pre-menstrual tension (distinguished from PMDD by milder symptoms and less impact on normal activities)" and from any co-existing psychiatric disorder.

DOSAGE AND METHOD OF ADMINISTRATION

For oral administration to adults only

Depression, with or without associated anxiety symptoms – adults and the elderly: A dose of 20mg/day is recommended.

Obsessive-compulsive disorder – adults and the elderly: 20mg/day to 60mg/day. A dose of 20mg/day is recommended as the initial dose. Although there may be an increased potential for side-effects at higher doses, a dose increase may be considered after several weeks if there is no response.

Pre-menstrual Dysphoric Disorder (PMDD): A dose of 20mg per day is recommended. Initial treatment should be limited to 6 months, after which patients should be reassessed regarding the benefit of continued therapy. All indications: The recommended dose may be increased or decreased. Doses above 80mg/day have not been systematically evaluated. Fluoxetine may be administered with or without food. When dosing is stopped, active drug substances will persist in the body for weeks. This should be borne in mind when starting or stopping treatment. Dosage tapering is unnecessary in most patients. Children: The use of fluoxetine in children is not recommended, as safety and efficacy have not been established. A lower or less frequent dose should be considered in patients with hepatic impairment, concurrent diseases (see 'Precautions'), or who are taking multiple medications.

CONTRA-INDICATIONS

Hypersensitivity to fluoxetine or to any of its excipients

Monoamine Oxidase Inhibitors: There have been reported serious and sometimes fatal reactions in patients receiving an SSRI in combination with a Mono Amine Oxidase Inhibitor (MAOI) and in patients who have been recently discontinued an SSRI and have been started on MAOI. Therefore the concomitant administration of the two drugs is contraindicated. In patients receiving MAOI fluoxetine should be administered at least 14 days after the discontinuation of an irreversible MAOI. Some cases presented with features resembling serotonin syndrome (which may resemble and be diagnosed as neuroleptic malignant syndrome).

Cyproheptadine or dantrolene may benefit patients experiencing such reactions. Symptoms of a drug interaction with a MAOI include: hyperthermia, rigidity, myoclonus, autonomic instability with possible rapid fluctuations of vital signs, mental status changes that include confusion, irritability and extreme agitation progressing to delirium and coma. When a MAOI must be administered in patients receiving fluoxetine at least a 5 week interval must be allowed before starting the administration of a MAOI. If fluoxetine has been prescribed chronically and/or at high dose, a longer interval should be considered.

The combination is not recommended. Treatment with fluoxetine can be initiated the following day after discontinuation of a reversible MAOI (e.g. moclobemide).

SPECIAL WARNINGS AND PRECAUTIONS FOR USE

Fluoxetine should be used with caution and in reduced doses in patients with impaired hepatic or renal function. Because of its epileptogenic effect, fluoxetine should be used with caution in patients with epilepsy or a history of such disorders. It should be avoided if the epilepsy is poorly controlled.

Caution is warranted in diabetic patients as fluoxetine may alter glycaemic control.

Fluoxetine should be discontinued in patients who develop a rash since systemic effects, possibly related to vasculitis, have occurred in such patients.

Suicide is an inherent risk in depressed patients; therefore patients should be closely monitored during early therapy until improvement in depression is observed. Fluoxetine should be used with caution in patients with a history of mania. It should be discontinued in patient entering as manic phase. Fluoxetine may impair performance of skilled tasks and if affected, patients should not drive or operate machinery. Fluoxetine is not recommended for nursing mothers as significant amounts of fluoxetine are distributed into breast milk. Fluoxetine should be withdrawn gradually to reduce the risk of withdrawal symptoms.

Suicidality in Children and Adolescents

Antidepressants increase the risk of suicidal thinking and behavior (suicidality) in children and adolescents with major depressive disorder (MOD) and other psychiatric disorders. Anyone considering the use of an antidepressant in a child or adolescent for any clinical use must balance the risk of increased suicidality with the clinical need. Patients who are started on therapy should be observed closely for clinical worsening, suicidality, or unusual changes in behaviour. Families and caregivers should be advised to closely observe the patient and to communicate with the prescriber.

INTERACTIONS WITH OTHER MEDICAMENTS AND OTHER FORMS OF INTERACTION

Half-life: The long elimination half-lives of both fluoxetine and norfluoxetine should be borne in mind (see section **Pharmacokinetic properties**) when considering pharmacodynamic or pharmacokinetic drug interactions (e.g. when switching from fluoxetine to other antidepressants).

Monoamine Oxidase inhibitors: See Contraindications

Not recommended combinations: MAOI-A (see section Contraindications).

Combinations requiring precautions for use: MAOI-B (selegiline): risk of serotonin syndrome. Clinical monitoring is recommended.

Phenytoin: Changes in blood levels have been observed when combined with fluoxetine. In some cases manifestations of toxicity have occurred. Consideration should be given to using conservative titration schedules of the concomitant drug and to monitoring clinical status.

Serotonergic drugs: Co-administration with serotonergic drugs (e.g. tramadol, triptans) may increase the risk of serotonin syndrome. Use with triptans carries the additional risk of coronary vasoconstriction and hypertension.

Lithium and tryptophan: There have been reports of serotonin syndrome when SSRIs have been given with lithium or tryptophan and, therefore, the concomitant use of fluoxetine with these drugs should be undertaken with caution. When fluoxetine is used in combination with lithium, closer and more frequent clinical monitoring is required.

CYP2D6 isoenzyme: Because fluoxetine metabolism (like tricyclic antidepressants and other selective serotonin antidepressants) involves the hepatic cytochrome CYP2D6 isoenzyme system, concomitant therapy with drugs also metabolised by this enzyme system may lead to drug interactions. Concomitant therapy with drugs predominantly metabolised by this isoenzyme, and which have a narrow therapeutic index (such as flecainide, encainide, carbamazepine, and tricyclic antidepressants), should be initiated at or adjusted to the low end of their dose range. This will also apply if fluoxetine has been taken in the previous 5 weeks.

Oral anticoagulants: Altered anticoagulant effects (laboratory values and/or clinical signs and symptoms), with no consistent pattern, but including increased bleeding, have been reported uncommonly when fluoxetine is co-administered with oral anticoagulants. Patients receiving warfarin therapy should receive careful coagulation monitoring when fluoxetine is initiated or stopped (see section Special warnings and precautions for use).

Electroconvulsive therapy (ECT): There have been rare reports of prolonged seizures in patients on fluoxetine receiving ECT treatment, therefore caution is advisable.

Alcohol: In formal testing, fluoxetine did not raise blood alcohol levels or enhance the effects of alcohol. However, the combination of SSRI treatment and alcohol is not advisable.

St John's Wort: In common with other SSRIs, pharmacodynamic interactions between fluoxetine and the herbal remedy St John's Wort (*Hypericum perforatum*) may occur, which may result in an increase of undesirable effects.

PREGNANCY AND LACTATION

There is limited experience of the use of Fluoxetine during pregnancy in humans, with suggested increase in incidence of miscarriages. No increase in the frequency of major foetal malformations has been recorded.

[Observational data indicate an increased risk \(less than 2-fold\) of postpartum haemorrhage following SSRI/SNRI exposure within the month prior to birth.](#)

Fluoxetine can be excreted into human breast milk and cause irritability in the infant.

Fluoxetine should not be prescribed during pregnancy and lactation unless considered essential by the physician.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

Although fluoxetine has been shown not to affect psychomotor performance in healthy volunteers, any psychoactive drug may impair judgment or skills. Patients should be advised to avoid driving a car or operating hazardous machinery until they are reasonably certain that their performance is not affected.

UNDESIRABLE EFFECTS

In common with other SSRIs, the following undesirable effects have been seen:

Body as a Whole: Hypersensitivity (e.g. pruritus, rash, urticaria, anaphylactoid reaction, vasculitis, serum sickness-like reaction, angioedema) (see Contraindications and Warnings), chills, serotonin syndrome, photosensitivity.

Digestive System: Gastrointestinal disorders (e.g. diarrhoea, nausea, vomiting, dyspepsia, dysphagia, taste perversion), dry mouth. Abnormal liver function tests have been reported rarely. Very rare cases of idiosyncratic hepatitis.

Nervous System: Headache, sleep abnormalities (eg, abnormal dreams, insomnia), dizziness, anorexia, fatigue (eg, somnolence, drowsiness), euphoria, transient abnormal movement (eg, twitching, ataxia, tremor, myoclonus), seizures and psychomotor restlessness. Hallucinations, manic reaction, confusion, agitation, anxiety and associated symptoms (e.g. nervousness), impaired concentration and thought process (eg, depersonalization), panic attacks (these symptoms may be due to the underlying disease).

Urogenital System: Urinary retention, urinary frequency.

Reproductive Disorders: Sexual dysfunction (delayed or absent ejaculation, anorgasm), priapism, galactorrhoea.

Miscellaneous: Alopecia, yawn, abnormal vision (e.g. blurred vision, mydriasis), ecchymosis (see Precautions), sweating, vasodilatation, arthralgia, myalgia, postural hypotension. **Hyponatraemia:** Hyponatraemia (including serum sodium <110 mmol/L) has been rarely reported and appeared to be reversible when Flosec was discontinued. Some cases were possibly due to the syndrome of inappropriate antidiuretic hormone secretion. The majority of reports were associated with older patients and patients taking diuretics or otherwise volume-depleted. Adverse effects may decrease in intensity and frequency with continued treatment and do not generally lead to cessation of therapy. When stopping treatment, withdrawal symptoms have been reported in association with selective serotonin re-uptake inhibitors (SSRI), although the available evidence does not suggest this is due to dependence. Common symptoms include dizziness, paraesthesia, headache, anxiety and nausea, the majority of which are mild and self-limiting. Fluoxetine has been only rarely associated with such symptoms. Plasma fluoxetine and norfluoxetine concentrations decrease gradually at the conclusion of therapy, which makes dosage tapering unnecessary in most patients.

OVERDOSE

Cases of overdose of fluoxetine alone usually have a mild course. Symptoms of overdose have included nausea, vomiting, seizures, cardiovascular dysfunction ranging from asymptomatic arrhythmias to cardiac arrest, pulmonary dysfunction, and signs of altered CNS status ranging from excitation to coma. Fatality attributed to overdose of fluoxetine alone has been extremely rare. Cardiac and vital signs monitoring are recommended, along with general symptomatic and supportive measures. No specific antidote is known.

Forced diuresis, dialysis, haemoperfusion, and exchange transfusion are unlikely to be of benefit. Activated charcoal, which may be used with sorbitol, may be as or more effective than emesis or lavage. In managing overdosage, consider the possibility of multiple drug involvement. An extended time for close medical observation may be needed in patients who have taken excessive quantities of a tricyclic antidepressant if they are also taking, or have recently taken, fluoxetine.

PHARMACOLOGICAL PROPERTIES

Pharmacodynamic Properties

Fluoxetine is a selective inhibitor of serotonin reuptake, and this probably accounts for the mechanism of action. Fluoxetine has practically no affinity to other receptors such as μ_1 -, μ_2 -and β -adrenergic; serotonergic; dopaminergic; histaminergic 1; muscarinic; and GABA receptors.

Pre-menstrual Dysphoric Disorder (PMDD): In clinical trials fluoxetine was shown to be effective in relieving both the cyclical mood changes and the physical symptoms (such as tension, irritability and dysphoria, bloating and breast tenderness) associated with PMDD. The aetiology of Pre-menstrual Dysphoric Disorder is unknown, but endogenous steroids (neuro and/or ovarian) involved in the menstrual cycle may interrelate with neuronal serotonergic activity.

Pharmacokinetic Properties

Absorption: Fluoxetine is well absorbed from the gastro-intestinal tract after oral administration. The bioavailability is not affected by food intake.

Distribution: Fluoxetine is extensively bound to plasma proteins (about 95%) and it is widely distributed (volume of distribution: 20-40 l/kg). Steady-state plasma concentrations are achieved after dosing for several weeks. Steady-state

concentrations after prolonged dosing are similar to concentrations seen at 4 to 5 weeks.

Metabolism: Fluoxetine has a non-linear pharmacokinetic profile with first pass liver effect. Maximum plasma concentration is generally achieved 6 to 8 hours after administration. Fluoxetine is extensively metabolised by the polymorphic enzyme CYP2D6. Fluoxetine is primarily metabolised by the liver to the active metabolite norfluoxetine (desmethylfluoxetine), by desmethylation.

Elimination: The elimination half-life of fluoxetine is 4 to 6 days and for norfluoxetine 4 to 16 days. These long half-lives are responsible for persistence of the drug for 5-6 weeks after discontinuation. Excretion is mainly (about 60%) via the kidney. Fluoxetine is secreted into breast milk.

At-risk populations

Elderly: Kinetic parameters are not altered in healthy elderly when compared to younger subjects.

Children and adolescents: The mean fluoxetine concentration in children is approximately 2-fold higher than that observed in adolescents and the mean norfluoxetine concentration 1.5-fold higher. Steady state plasma concentrations are dependent on body weight and are higher in lower weight children (see Posology and method of administration). As in adults, fluoxetine and norfluoxetine accumulated extensively following multiple oral dosing; steady-state concentrations were achieved within 3 to 4 weeks of daily dosing.

Hepatic insufficiency: In case of hepatic insufficiency (alcoholic cirrhosis), fluoxetine and norfluoxetine half-lives are increased to 7 and 12 days, respectively. A lower or less frequent dose should be considered.

Renal insufficiency: After single-dose administration of fluoxetine in patients with mild, moderate, or complete (anuria) renal insufficiency, kinetic parameters have not been altered when compared to healthy volunteers. However, after repeated administration, an increase in steady-state plateau of plasma concentrations may be observed.

PHARMACEUTICAL PARTICULARS

List of Excipients

Violet/Yellow capsules: lactose monohydrate, cellulose microcrystalline, magnesium stearate, titanium oxide (E171), gelatin, erythrosine (E127), Iron oxide yellow (E172), Iron oxide black (E172), Indigotine (E132).

Incompatibilities

None

Shelf Life

48 months from the date of production

Special Precautions for Storage

Store below 30°C in the original package

Nature and Contents of Container

MAGRILAN capsules 20mg are packed in Aluminum-PVC blisters available in packs of 10, 12, 20, 30 and 100 capsules.

Instruction for Use/Handling

None

NAME AND ADDRESS OF MANUFACTURER

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